Syed Jawad Ali

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Technical Skills

Programming: Python, JavaScript, TypeScript
Frameworks: Django, FastAPI, Next.js
Databases: PostgreSQL, MySQL, SQLite, pgvector
Generative AI & LLMs: Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, LangGraph, Agentic AI and Multi-Agent Systems, Hallucination Mitigation
Retrieval & Vector Systems: ChromaDB, PostgreSQL Vector Store (pgvector), Hybrid Retrieval (Keyword + Vector), Embeddings, Similarity Search
Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, Pandas, NumPy, Reinforcement Learning (DQN)
Systems & Tools: Docker, Git, Redis, REST APIs, Scalable System Design

Publications

RAGENT: DQN-Based Multi-Agent RAG System
Sabina Akhar, Syed Jawad Ali
Submitted to Springer (Under Review)
DQN-based multi-agent framework for adaptive retrieval in RAG systems, improving factuality, reducing hallucinations, and optimizing retrieval efficiency.

DQN-RAG: Reinforcement Learning for Adaptive Retrieval in Multi-Hop QA
Syed Jawad Ali
Under Review / Ready for Submission
Reinforcement learning-based RAG system that dynamically selects retrieval and reasoning actions, improving accuracy and reducing token usage on multi-hop QA tasks.

Experience

Joyn Digital, LiftUpServices March 2024 – Present
Freelance Python Developer (Project-Based — Django, FastAPI, GenAI – LLM/RAG — Remote) *Islamabad, Pakistan*

- Designed and developed scalable backend systems using FastAPI and Django REST Framework, serving AI-powered applications.
- Built production-grade RAG (Retrieval-Augmented Generation) systems for intelligent document querying with citation support. Developed agentic AI pipelines integrating LLMs, enabling automated reasoning, task planning, and multi-step decision workflows.
- Integrated LLMs (BERT, Transformers, OpenAI APIs) with backend services to deliver real-time AI features. Implemented vector search systems (embeddings, similarity search) using PostgreSQL and optimized retrieval performance. Collaborated with cross-functional teams to deploy and scale AI services in production environments.

BUIC Research Lab April 2018 – May 2020
Machine Learning Engineer *Islamabad, Pakistan*

- Hands-on experience with machine learning libraries including Pandas, NumPy, Matplotlib, scikit-learn, Keras, and TensorFlow.
- Applied Machine Learning techniques including Supervised and Unsupervised algorithms for Classification, Clustering, and Dimensionality Reduction, as well as NLP using the Bidirectional Encoder Representations from Transformers (BERT) model and CNNs.

NextIn Technologies August 2016 – Dec 2018
Dot Net Developer *Islamabad, Pakistan*

- Worked as ASP.NET Web Form and MVC developer.
- Transformed business logic into scalable backend solutions and collaborated with cross-functional teams to design, implement, and optimize systems.

Projects

AI-Powered Multi-Agent Retrieval Platform | *Python, Django, LangGraph, PyTorch, PostgreSQL, Redis, Docker*

- Developed a scalable Python-based Retrieval-Augmented Generation (RAG) platform integrating multiple AI agents for document retrieval, web search, response evaluation, and answer generation.
- Built RESTful APIs using Django and implemented asynchronous task processing with Celery and Redis to improve system responsiveness and throughput.
- Designed and optimized database architectures using PostgreSQL and ChromaDB for efficient storage, indexing, and retrieval of structured and vector data.
- Containerized services with Docker and automated deployment workflows, enabling reliable and production-ready application delivery.

RAGENT: Intelligent AI Chatbot Platform | *Python, Django, PostgreSQL, VectorDB, Next.js, JWT*

- Developed a full-stack AI chatbot platform using Django and Next.js, supporting document-based knowledge retrieval and real-time web search integration.
- Designed and implemented secure REST APIs with JWT authentication, OTP verification, password recovery, and role-based access control.
- Integrated vector databases and retrieval pipelines to enable efficient semantic search and context-aware response generation.
- Implemented scalable backend services, database optimization, and asynchronous processing to support high-performance enterprise applications.

Intelligent RAG SaaS Platform (AI Knowledge Assistant) | *Python, Django, DRF, PostgreSQL, pgvector, Next.js, Docker*

- Developed a production-grade SaaS platform enabling organizations to query internal documents (PDFs, CSVs, and knowledge bases) using natural language interactions.
- Implemented hybrid search (vector + keyword) using PostgreSQL and pgvector, significantly improving retrieval accuracy and relevance.
- Designed context-aware response generation with source citations, query rewriting, and re-ranking mechanisms to enhance answer reliability.
- Built secure role-based access control (RBAC), conversation memory, user management, and administrative dashboards for enterprise deployment.
- Optimized backend performance through caching, database tuning, and scalable API design using Django REST Framework.

Multi-Agent AI System with Decision Intelligence (RAG + RL) | *Python, Django, LangGraph, PostgreSQL, Redis, Next.js*

- Designed and implemented a multi-agent AI architecture consisting of Planner, Retrieval, Evaluation, and Response agents for complex task execution.
- Developed a reinforcement learning–based decision engine to dynamically determine retrieval, response generation, or clarification actions based on query context.
- Implemented a state-action-reward framework to improve answer accuracy, reduce hallucinations, and optimize API usage costs.
- Built adaptive retrieval pipelines with re-ranking, confidence scoring, and iterative reasoning workflows for enhanced response quality.
- Developed secure backend services with RBAC, conversation memory, REST APIs, and administrative monitoring features.

AI-Powered Learning Management System (Agentic RAG LMS) | *Python, Django, PostgreSQL, pgvector, ChromaDB, Next.js*

- Developed a full-featured Learning Management System (LMS) supporting academic management, attendance, examinations, assignments, communication, reporting, and notifications.
- Integrated an Agentic RAG framework enabling intelligent question answering over course materials, institutional knowledge, and academic records.
- Designed semantic search infrastructure using PostgreSQL (pgvector) and ChromaDB for efficient embedding storage and retrieval.
- Implemented hybrid retrieval pipelines combining vector search and structured database queries to improve contextual relevance and answer accuracy.
- Built a multi-agent architecture for planning, retrieval, and response generation, enabling natural-language interaction with academic data and workflows.

AI-Powered Resume Screening and Candidate Matching System | *Python, Django, Next.js, NLP, PostgreSQL*

- Developed an AI-driven resume screening system to automatically analyze, rank, and match candidates against job descriptions.
- Implemented NLP-based semantic matching using embeddings to evaluate skill alignment, experience relevance, and job fit scoring.
- Designed explainable AI scoring mechanisms providing transparent insights and recommendations to support HR decision-making.
- Built scalable backend services using Django, enabling efficient processing of large-scale resume datasets and job postings.

Human Resource Management System (HRMS) | *Python, Django REST Framework, FastAPI, Next.js, PostgreSQL*

- Developed a scalable HR management platform for employee records, attendance tracking, and organizational data management.
- Built RESTful backend services using Django REST Framework and FastAPI, ensuring modular architecture and high-performance API responses.
- Integrated machine learning models to predict employee salaries based on experience and education, and to evaluate performance using attendance and productivity metrics.
- Developed a responsive frontend using Next.js, providing an efficient and modern user interface for HR operations.

Scalable E-Commerce Platform (Product, Order & Inventory System) | *Python, Django, PostgreSQL, Next.js, Redis*

- Developed a full-stack e-commerce platform supporting product management, categories, variants (size, color, material), and advanced filtering.
- Designed a real-time inventory management system with stock tracking, reorder levels, and automated SKU generation for product variants.
- Built order management system including cart, checkout flow, payment processing, and delivery tracking with multiple payment methods.
- Implemented sales, discount, and dynamic pricing modules with automated price calculation and promotional rules engine.

Scalable Blogging Platform with CMS Architecture | *Python, Django, PostgreSQL, Next.js*

- Developed a modular blogging platform with content management features including posts, categories, tags, media handling, comments, newsletters, and campaigns.
- Designed SEO-optimized content models with structured metadata, hierarchical categorization, and rich text editing for improved discoverability.
- Built a secure comment system with moderation workflows, nested replies, rate limiting, and role-based access control.
- Integrated newsletter and media management system with subscriber segmentation, scheduling, and automated content distribution.

Education

Bahria University Islamabad, Pakistan

Master of Science in Computer Science (CGPA: 3.40) *2020*

- **Relevant Coursework:** Data Science, Machine Learning, Deep Learning, and AI-based Thesis
- **Relevant Research:** Selection of Web Services using Artificial Intelligence

Federal Urdu University of Science and Arts Islamabad, Pakistan

Bachelor of Science in Computer Science (CGPA: 3.01) *2016*

- **Relevant Coursework:** Data Structures and Algorithms (C++), Programming Fundamentals (C++, Python), Web and Mobile Applications